

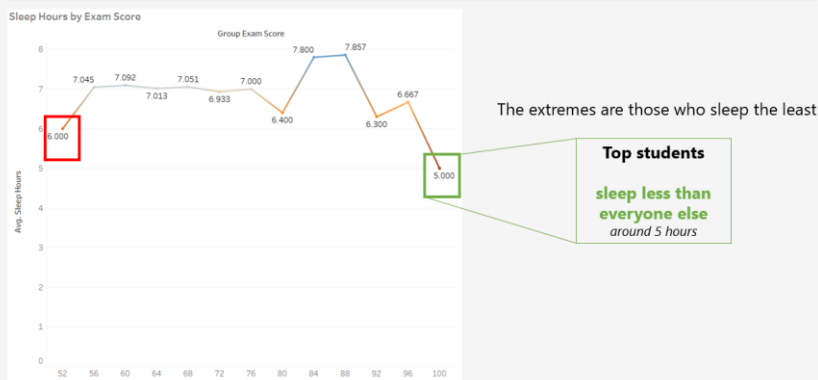


19 Factors
3 Approaches
1 Winner

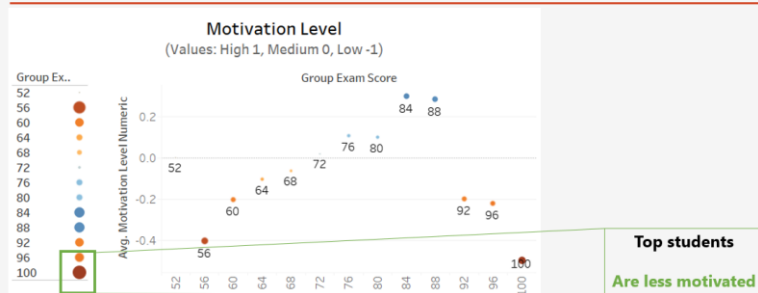
Which factor has the greatest impact on
students' performance?

Introduction

You snooze, you lose



Something Strange



1 Goal And Data Distribution

2 Why 3 Approaches?

3 Numerical Factors

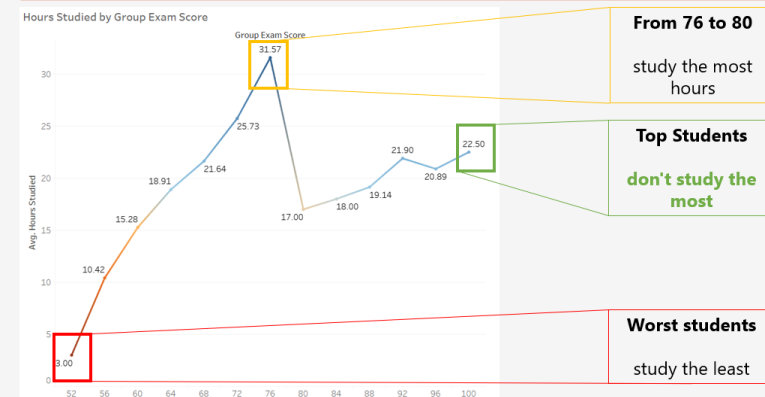
4 Binary Factors

5 Categorical Factors

6 Gender vs Some Factors

7 Winner

Quality over quantity

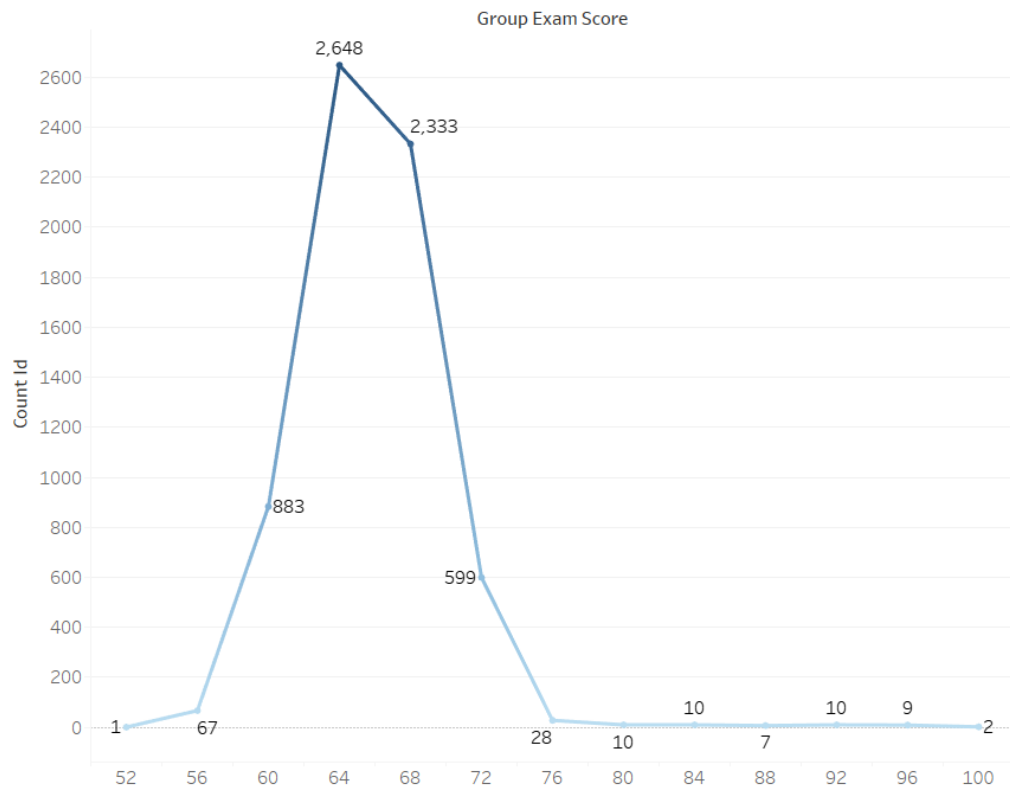


A well-rounded person is a successful person



Goal And Data Distribution

Data Distribution



Group Exam Score

- ☒ (All)
- ☒ 52
- ☒ 56
- ☒ 60
- ☒ 64
- ☒ 68
- ☒ 72
- ☒ 76
- ☒ 80
- ☒ 84
- ☒ 88
- ☒ 92
- ☒ 96
- ☒ 100

Score Parameter

4

AGG(Count Id)

1 2,648

The **goal** of the analysis is to determine which of the **19 factors** has the greatest **linear** impact on student's grade.

Then, with a correlation matrix, the factor that has the most influence will be determined.

(we need to consider the data distribution. As there are only a few students with high or low scores, which means we have limited data for those values)

Why 3 Approaches?

1 Binary Factors

- Gender (Male/Female)
- Learning Disabilities (Yes/No)
- Internet Access(Yes/No)
- School Type(Private/Public)
- Extracurricular Activities (Yes/No)

2 Categorical Factors

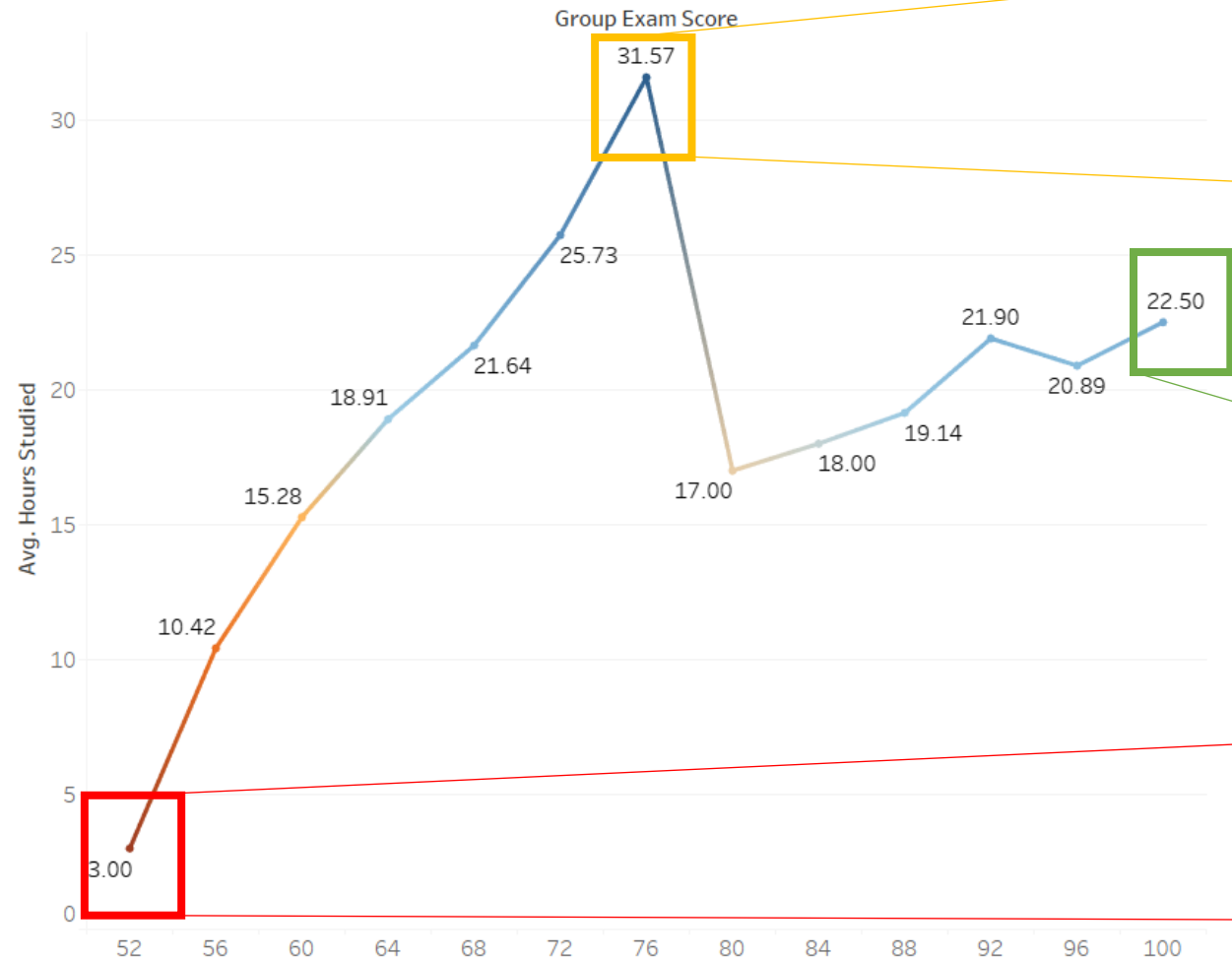
- Access to Resources (High/Medium/Low)
- Distance from Home (Far/Moderate/Near)
- Family Income (High/Medium/Low)
- Parental Involvement (High/Medium/Low)
- Parental Education Level (College/High School/Postgraduate)
- Motivation Level (High/Medium/Low)
- Peer Influence (Negative/ Neutral/Positive)
- Teacher Quality(High/Medium/Low)

3 Numerical Factors

- Attendance
- Physical Activity
- Hours Studied
- Previous Scores
- Sleep Hours
- Tutoring Sessions

Quality over quantity

Hours Studied by Group Exam Score



From 76 to 80

study the most hours

Top Students

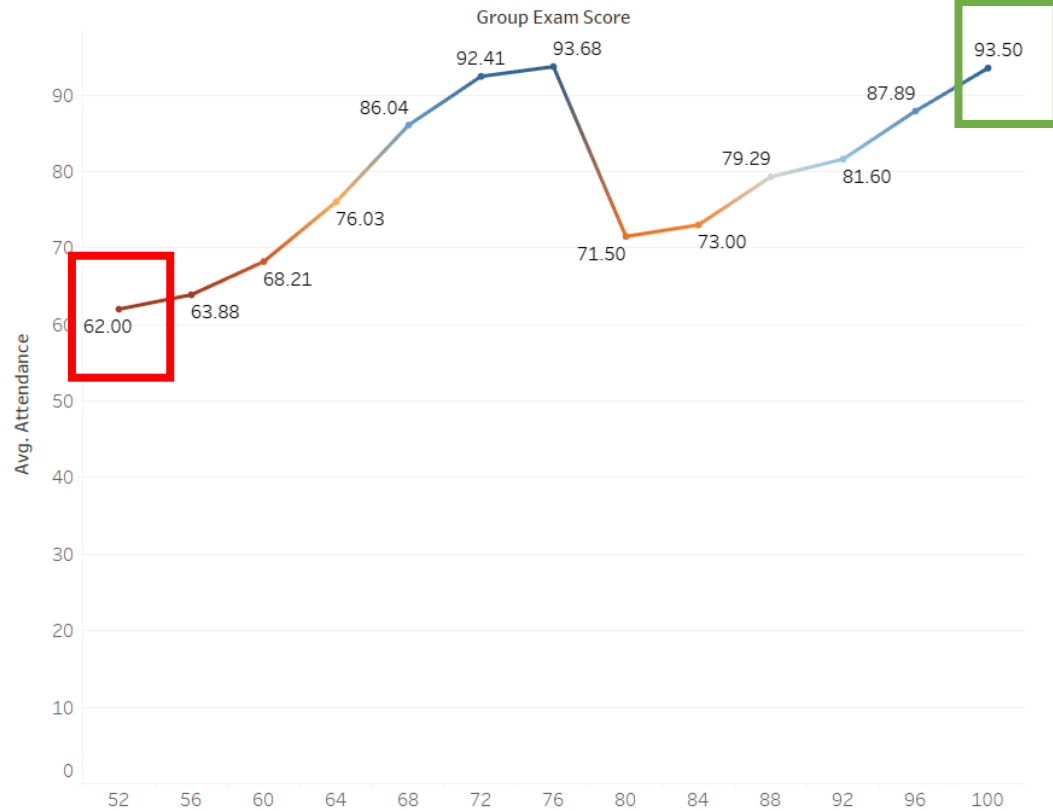
don't study the most

Worst students

study the least

Attending lessons and tutoring are essential

Attendance by Group Exam Score

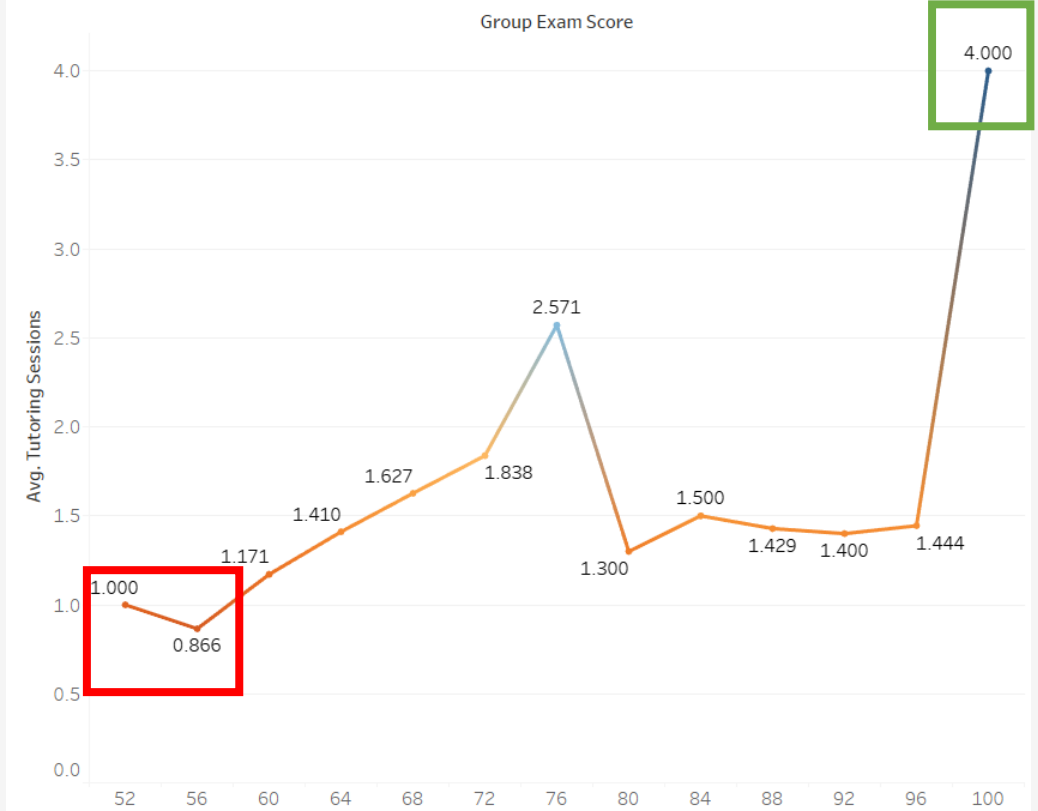


Worst students

- *The lowest attendance*
- *The fewest tutoring sessions*

VS

Tutoring Session by Group Exam Score

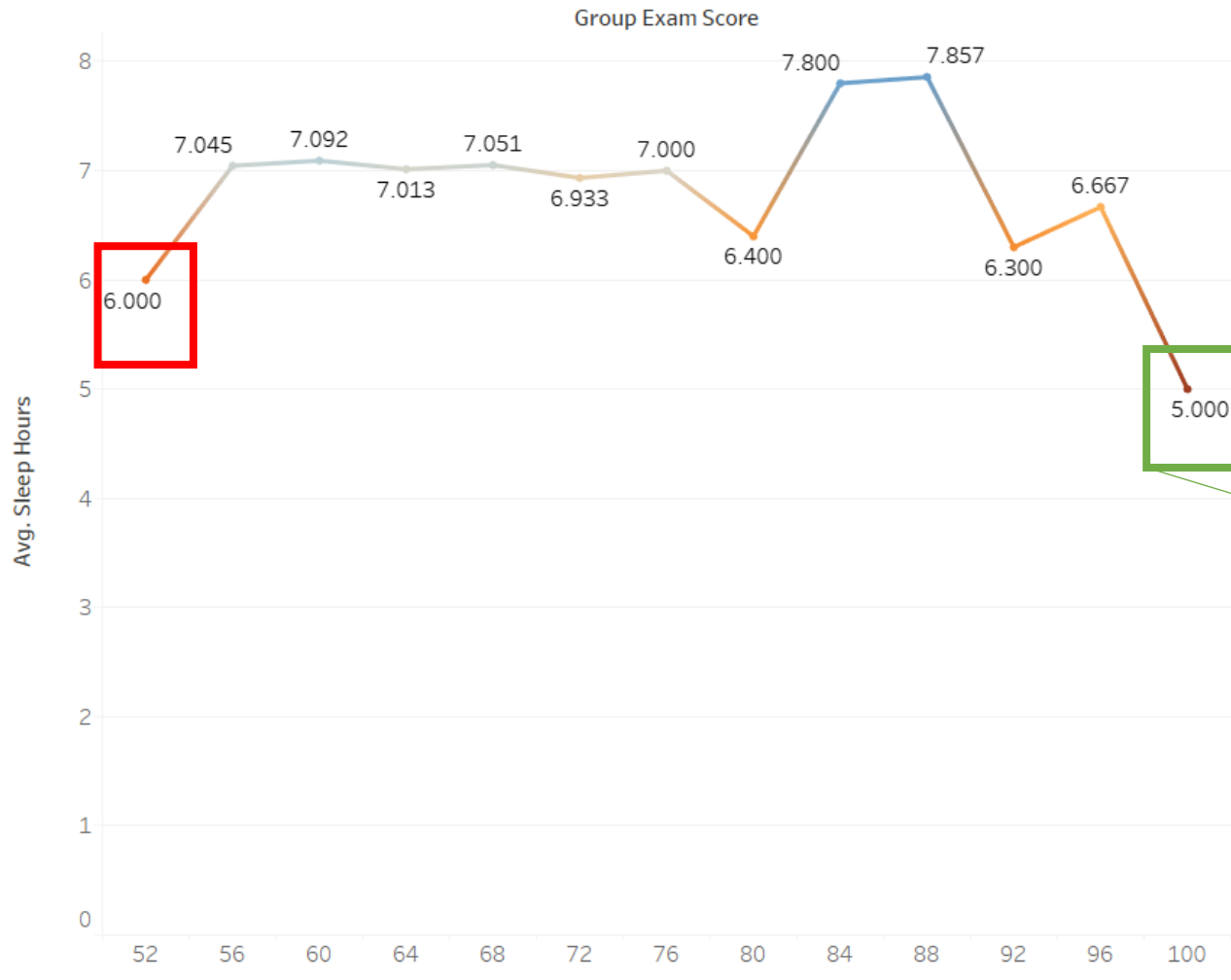


Top Students

- *High attendance*
- *The most tutoring sessions*

You snooze, you lose

Sleep Hours by Exam Score



The students with the extreme grades are those who sleep the least

Top students

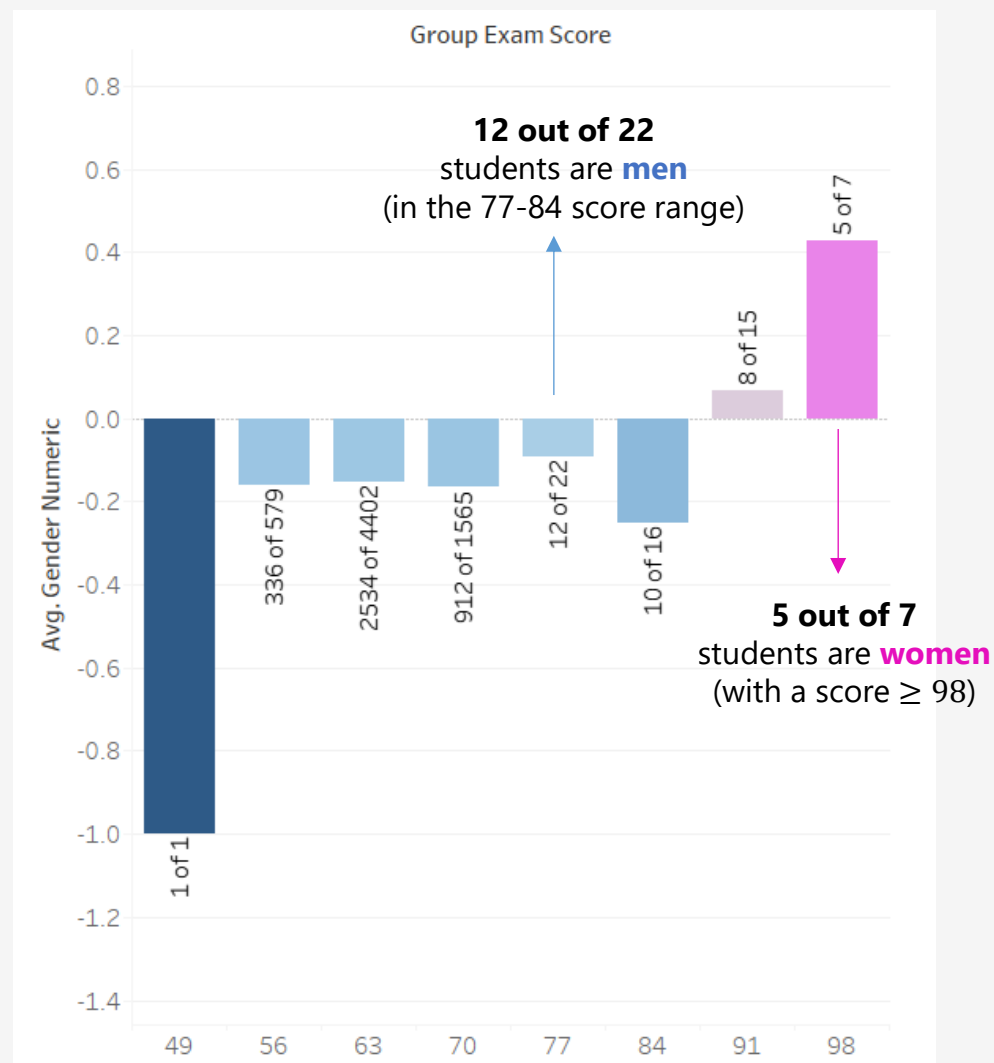
**sleep less than
everyone else**
around 5 hours

Possible Candidate

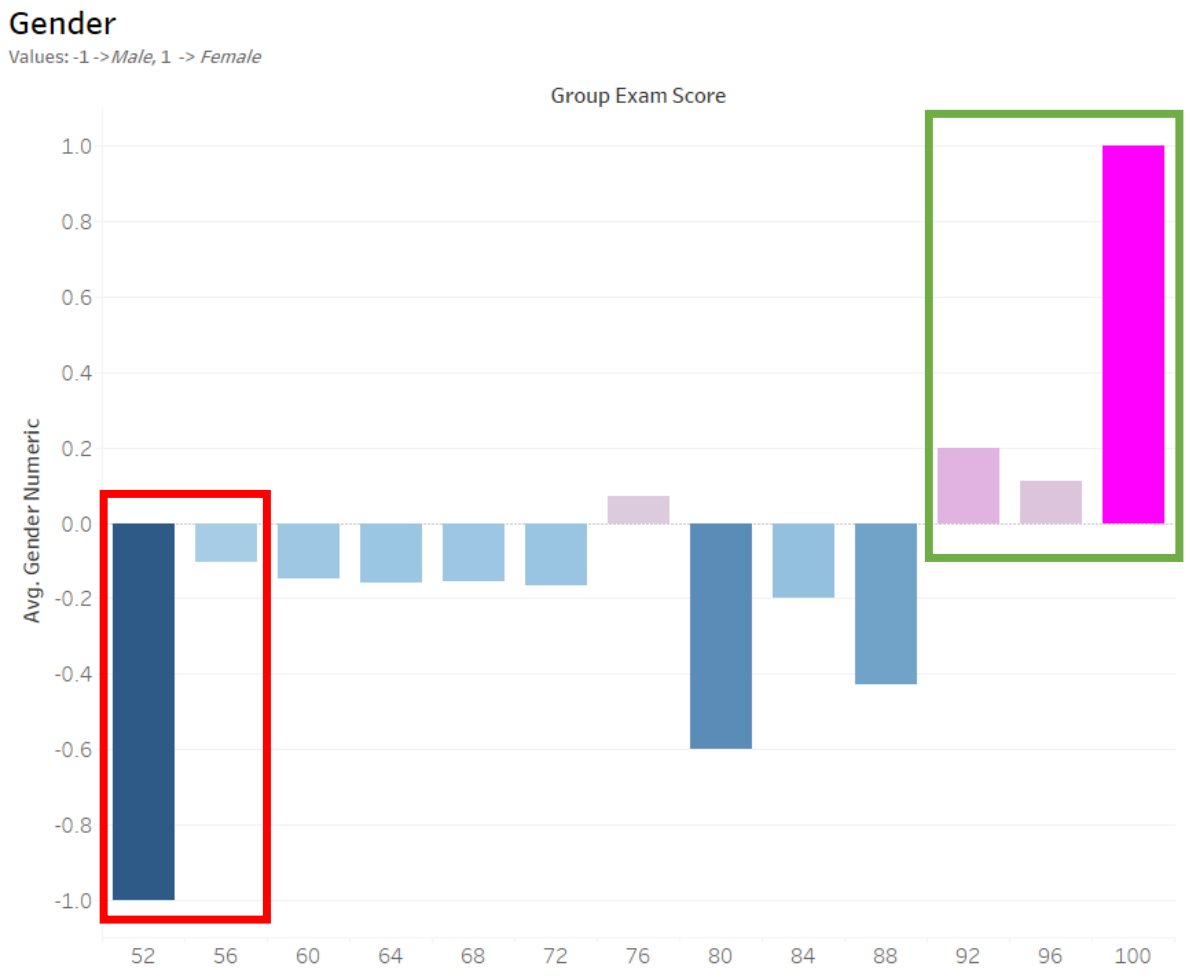


Binary Factors

- **1 or -1:** 100% of that grade group belongs to that category.
- **Otherwise:** it represents the percentage difference

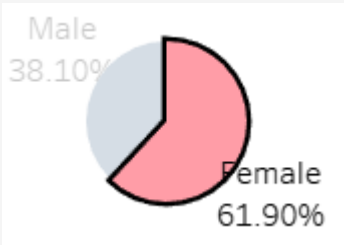


Nothing We Can Do About It



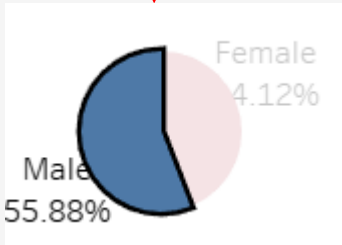
Top Students ≥ 92

*Most are **women***

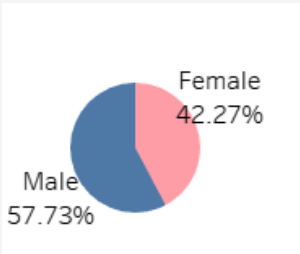


Worst Students < 60

*Most are **men***



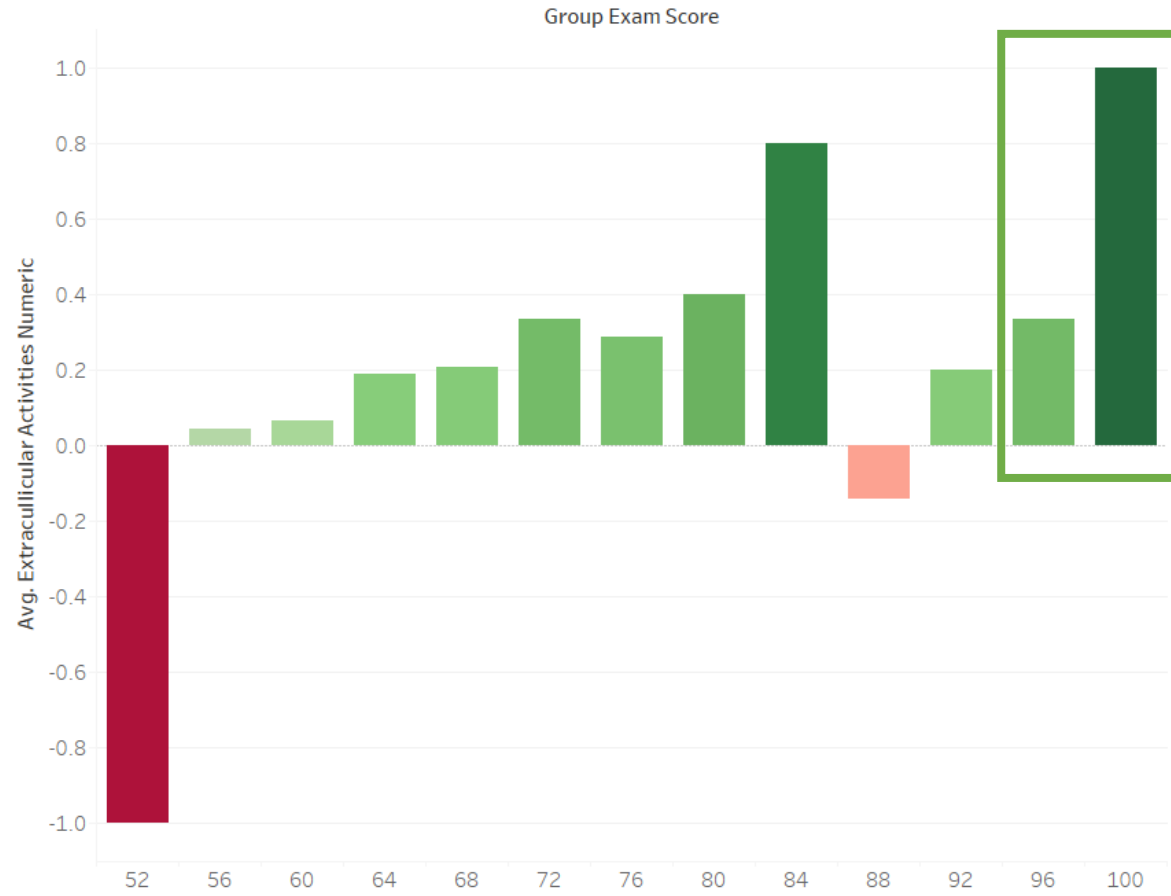
Considering also that women are fewer in the database, they are mainly concentrated in the higher score range



A well-rounded person is a successful person

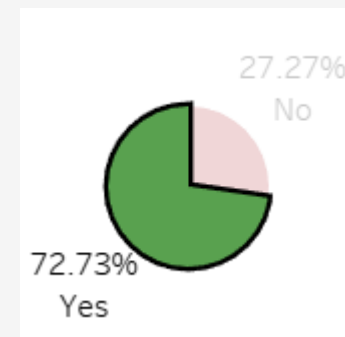
Extracurricular Activity

Values : -1-> No, 1-> Yes



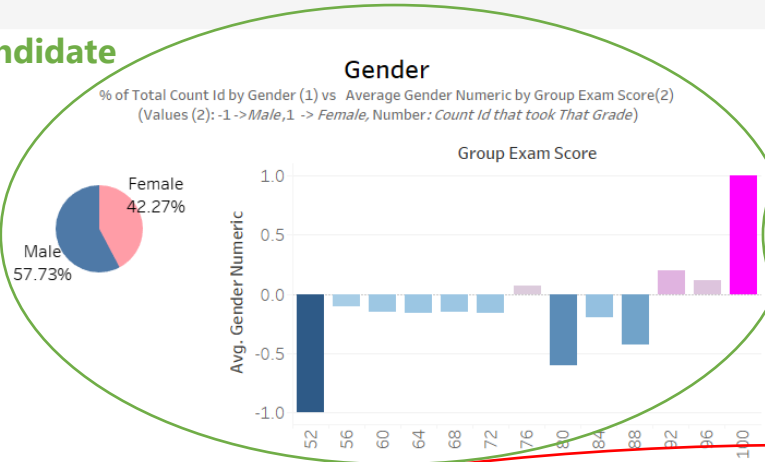
Top Students ≥ 96

*Most participate in
extracurricular
activities*

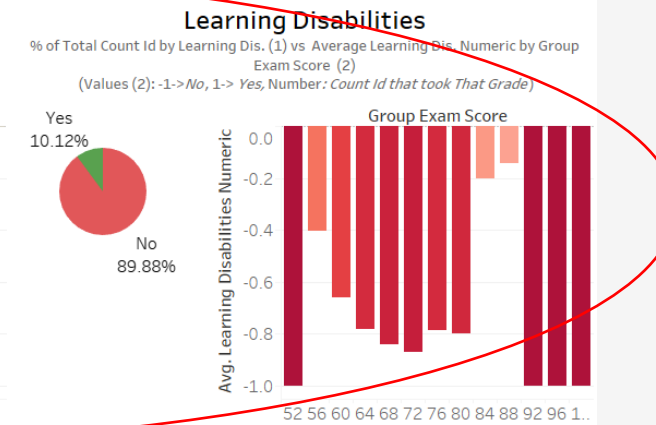
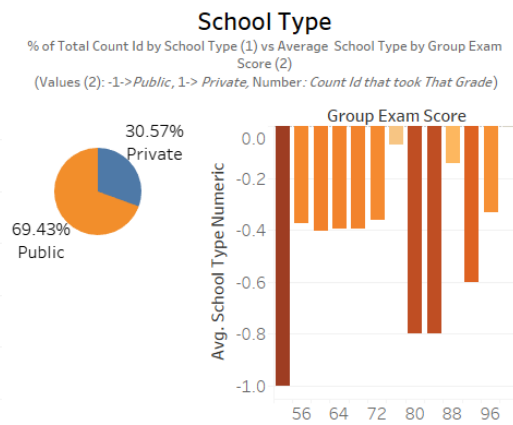
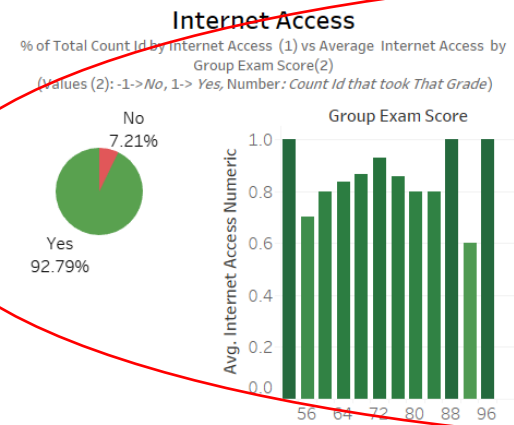
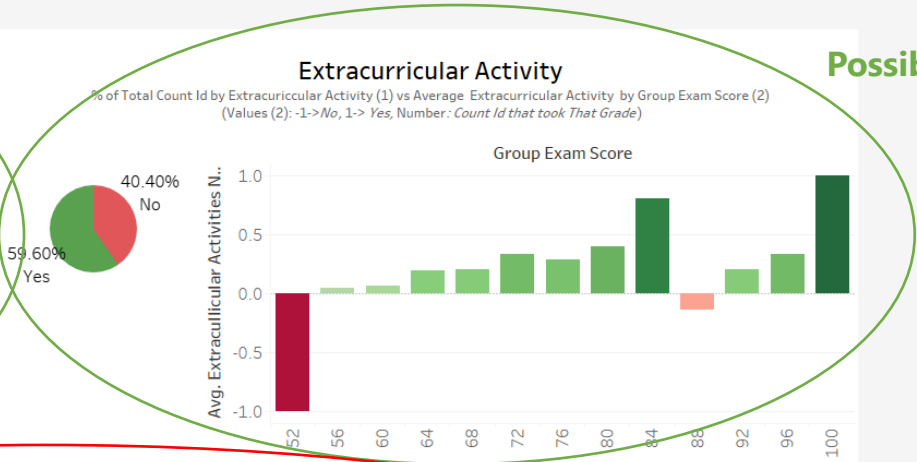


Back To the Goal

Possible Candidate



Possible Candidate



the database is primarily concentrated in one of the two categories

Categorical Factors

The two graphs represent Factor by grade group.

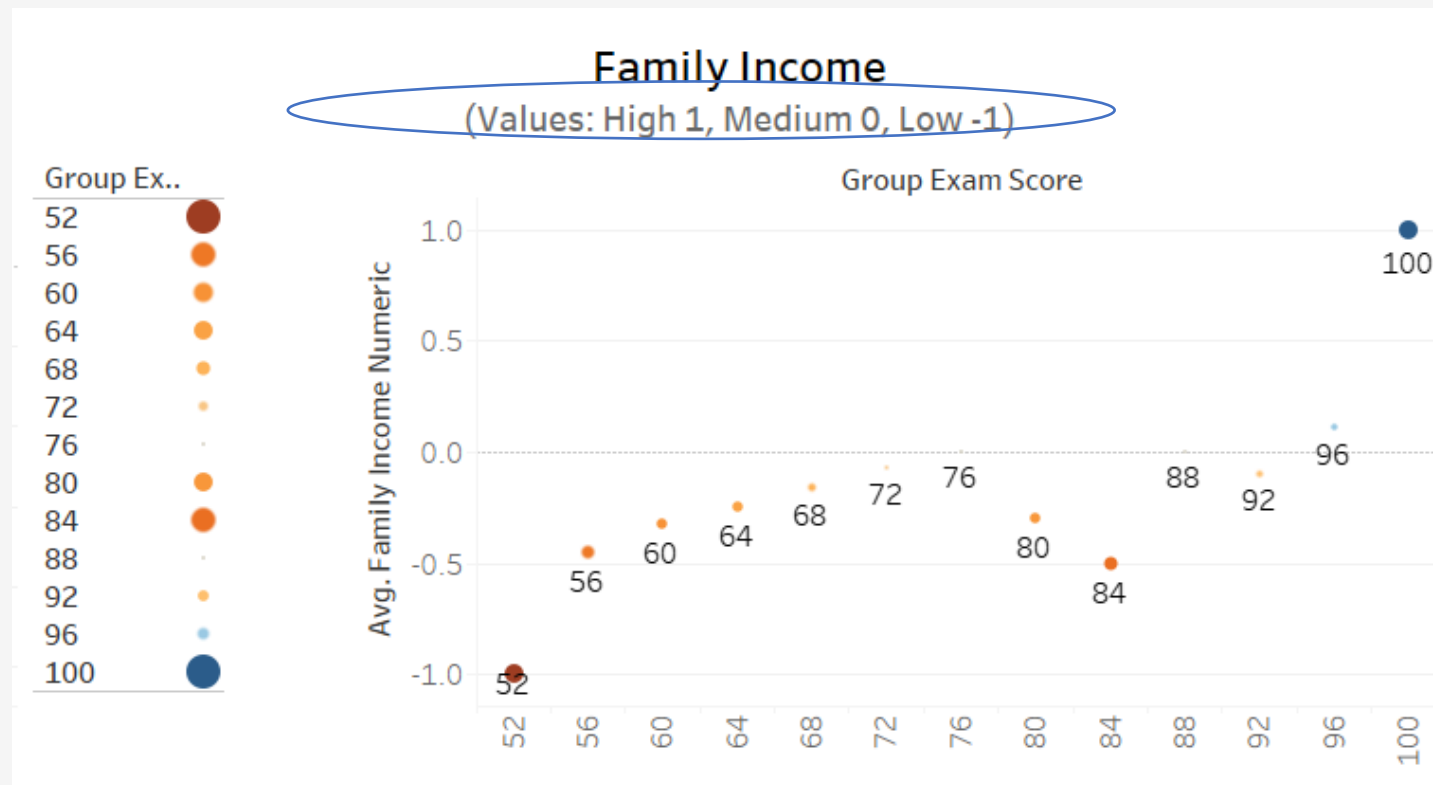
EXTREMES:

- A larger blue circle (value: 1) indicates that all students with that score fall into the "1 category".
- A larger red circle (value: -1) indicates the same for the "-1 category".

MIDDLE:

The closer the value is to 0:

- the more likely most students with that score are in the "0 category".
- The average between the extreme categories results in this outcome, which is **not very significant**.

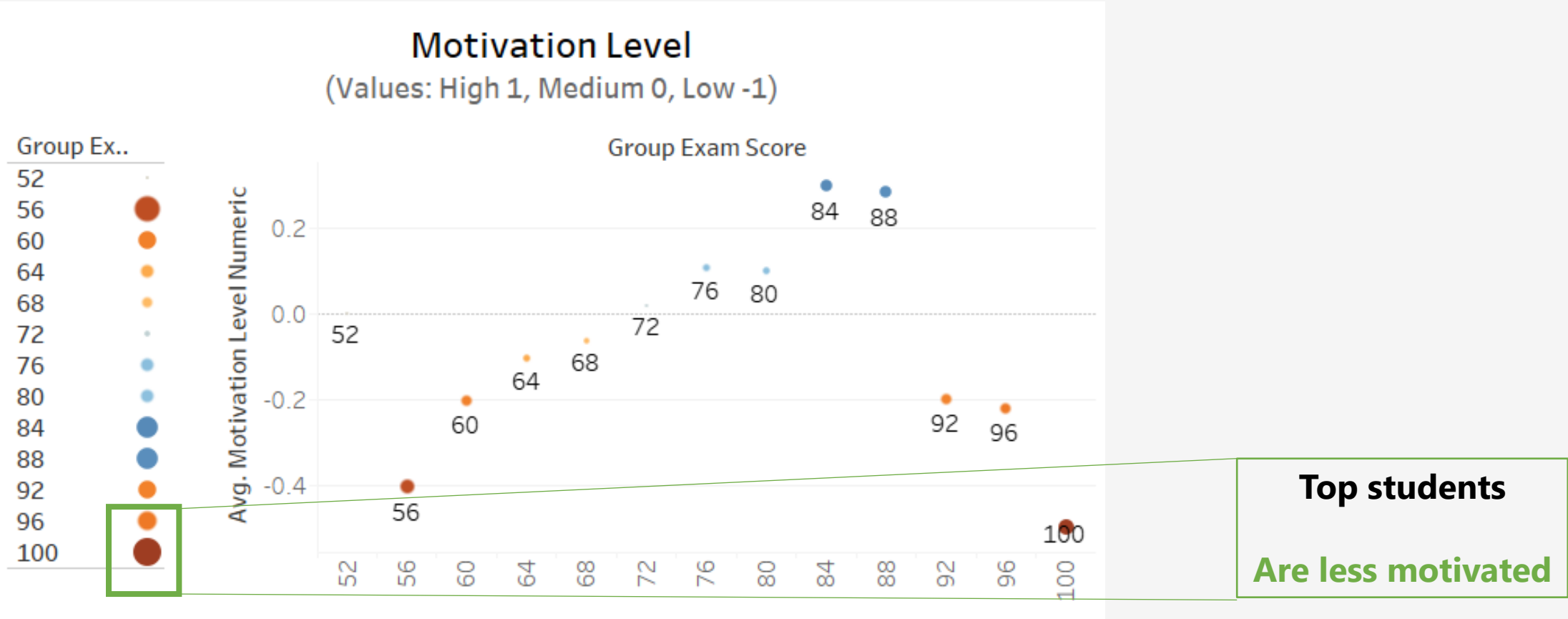


Why these graphs?

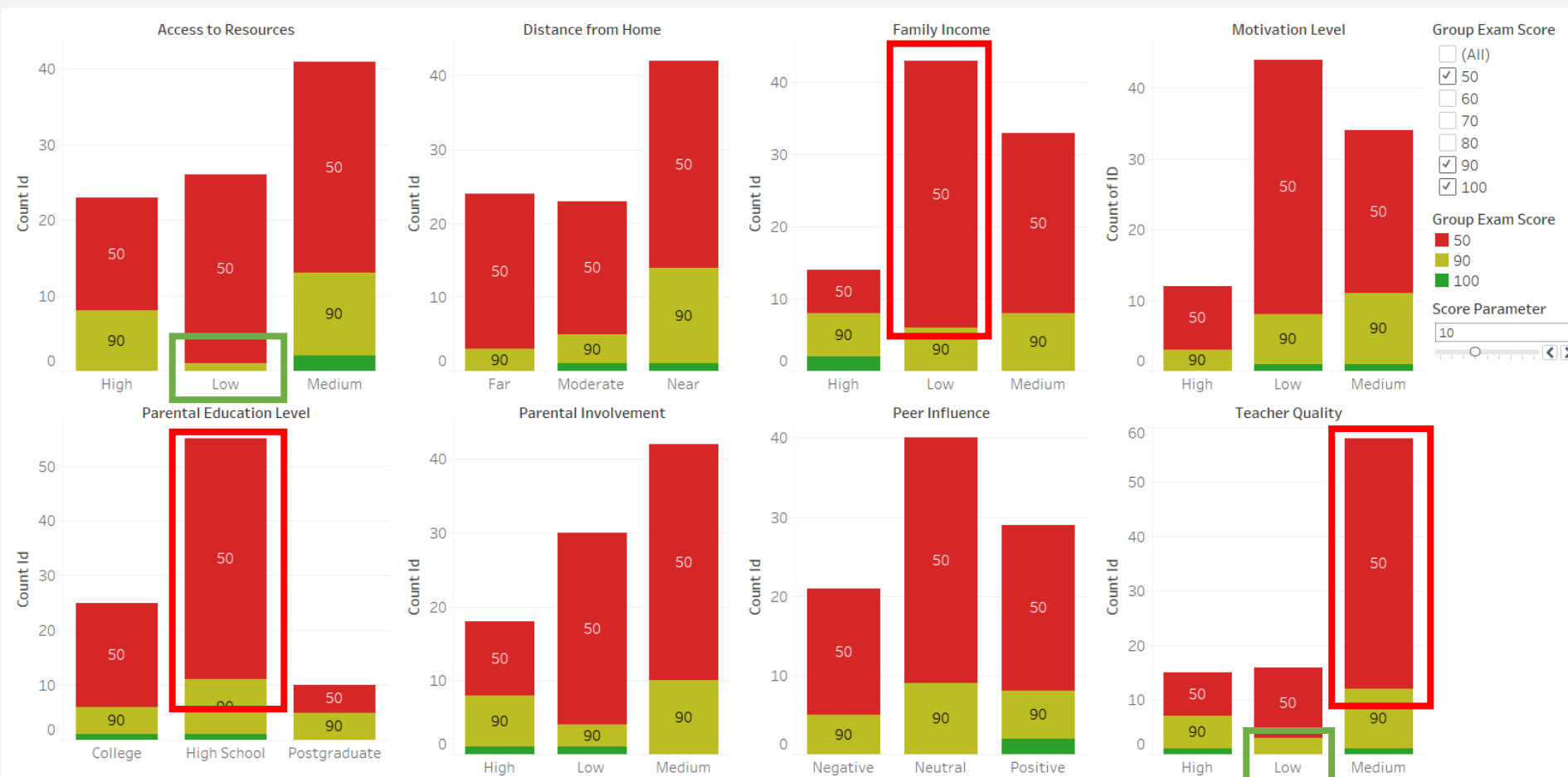
The goal of the analysis is to visualize a possible linear relationship with the Exam Score, allowing us to observe the trend as the scores increase, focusing on the "1 category" and "-1 category".

(Later, another type of graph will consider all categories equally.)

Something Strange



Focus: Top and Worst



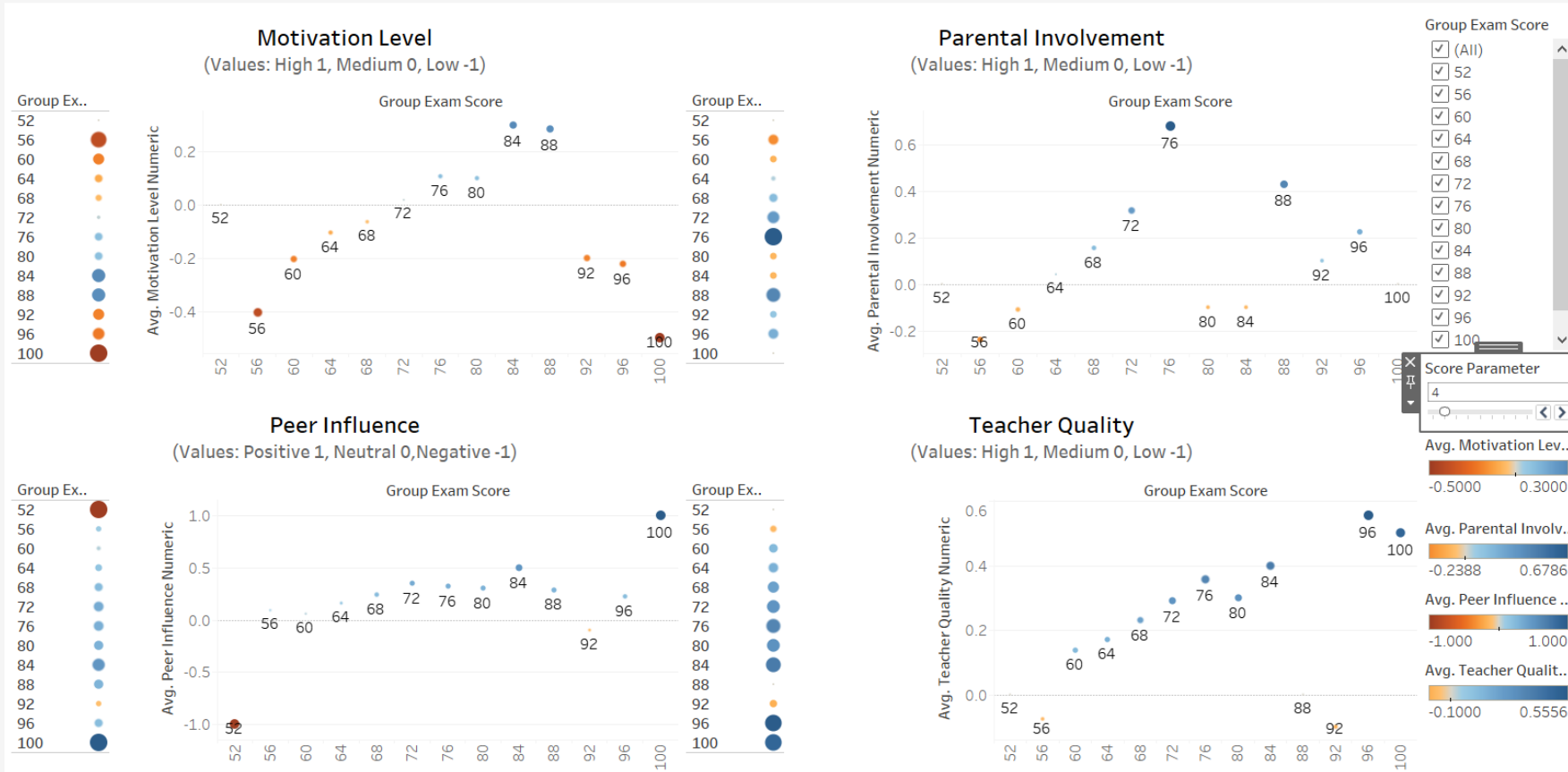
Top Students ≥ 90

They are fairly heterogeneous, except for **access to resources**, **teacher quality**. Most fall into the high and medium categories, for both factors, **but not very significant**.

Worst Students < 60

They are fairly heterogeneous, except for: **family income**, primarily low; **parental education**, mainly high school; and **teacher quality**, primarily medium

Back To the Goal

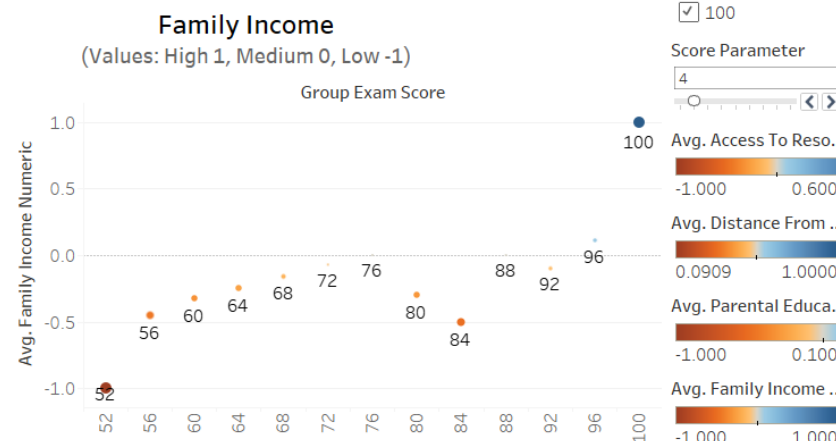
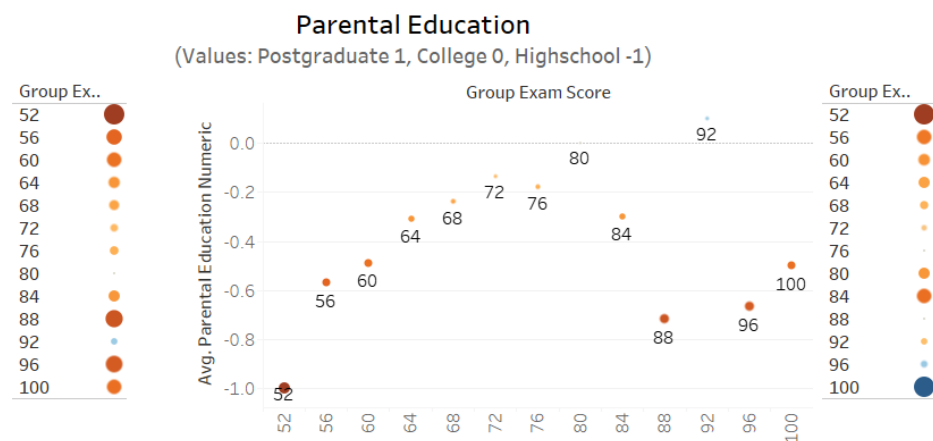
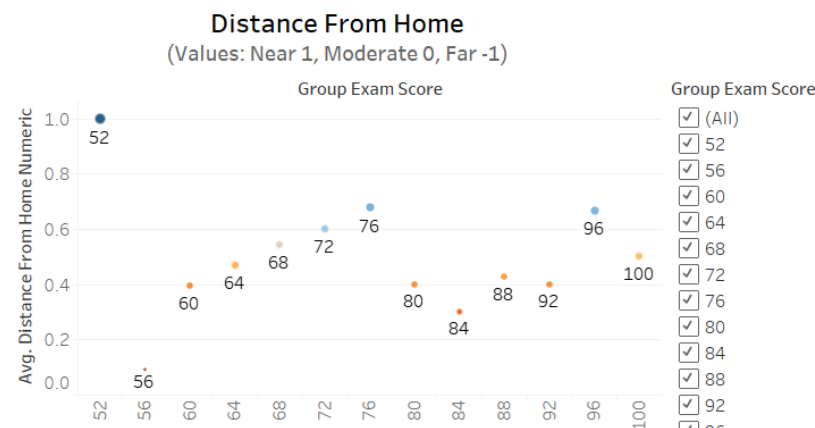
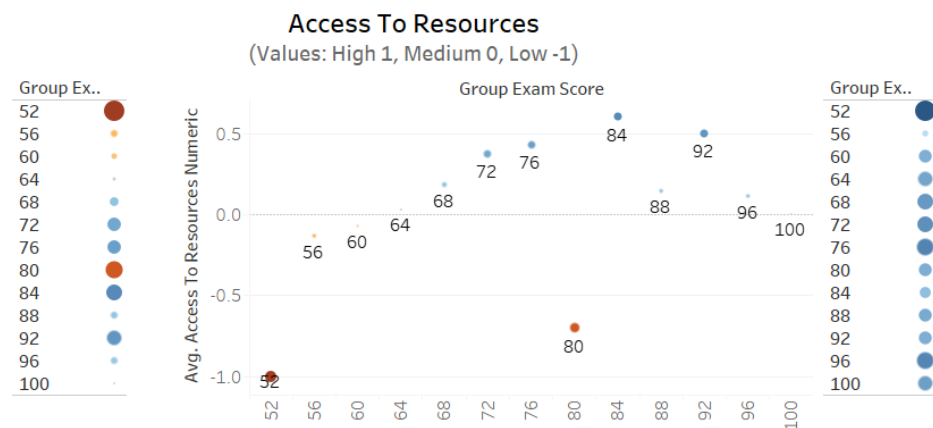


- **Parental involvement:** Those with higher parental involvement tend to have intermediate grades.
- **Peer influence:** Most students are in the middle.
- **Teacher quality:** the highest grades can be associated with students who had, on average, more qualified teachers.

No candidate: there doesn't seem to be a pattern

There are too few students with the grades ≥ 100 and < 56 to make any assumptions

Back To the Goal



- **Access to resources:** A different behavior can be observed in students who scored low and those who scored between 80-84, as they tend to have low access to resources.
- **Distance from home:** is fairly homogeneous in the dataset.
- **Parental education:** For almost everyone, it's negative, meaning high school level.
- **Family income:** It can be observed that there is a difference between the lowest and highest grades.

No candidates: there doesn't seem to be a pattern and the majority is in the same category

There are too few students with the grades ≥ 100 and < 56 to make any assumptions

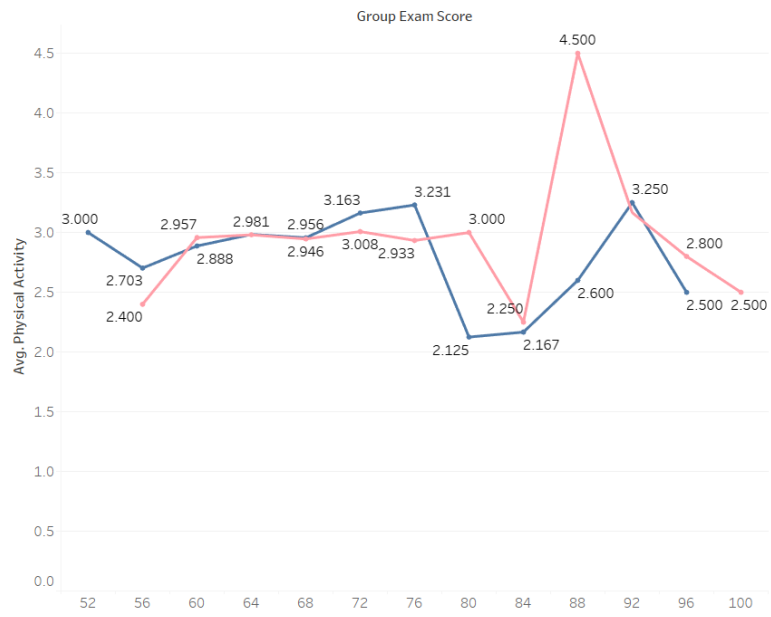
Gender by some Factors

Attendance by Group Exam Score with Gender



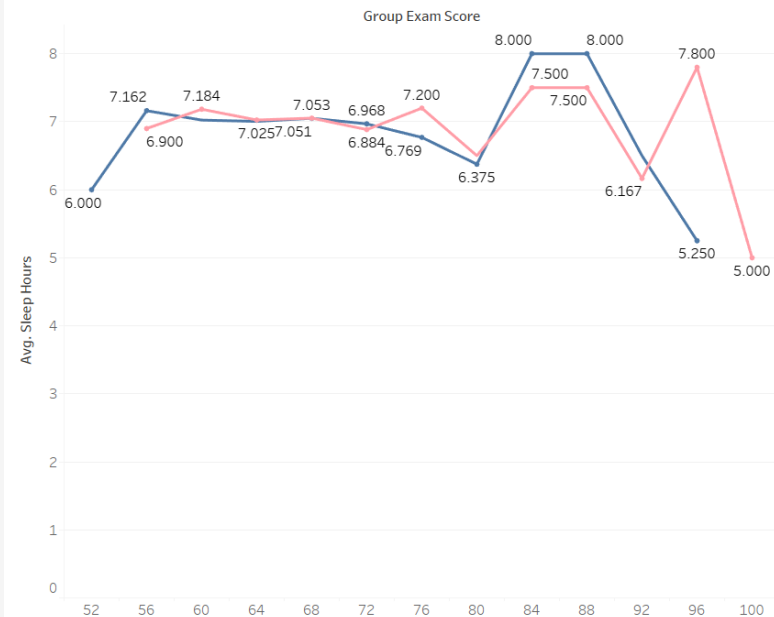
There isn't much difference in **attendance** between men and women

Physical Activity by Group Exam Score with Gender



Women
tend to exercise more

Sleep Hours by Exam Score with Gender



There isn't much difference in **sleep hours** between men and women

Who do you think is the winner?

1 Attendance

2 **Tutoring Sessions**

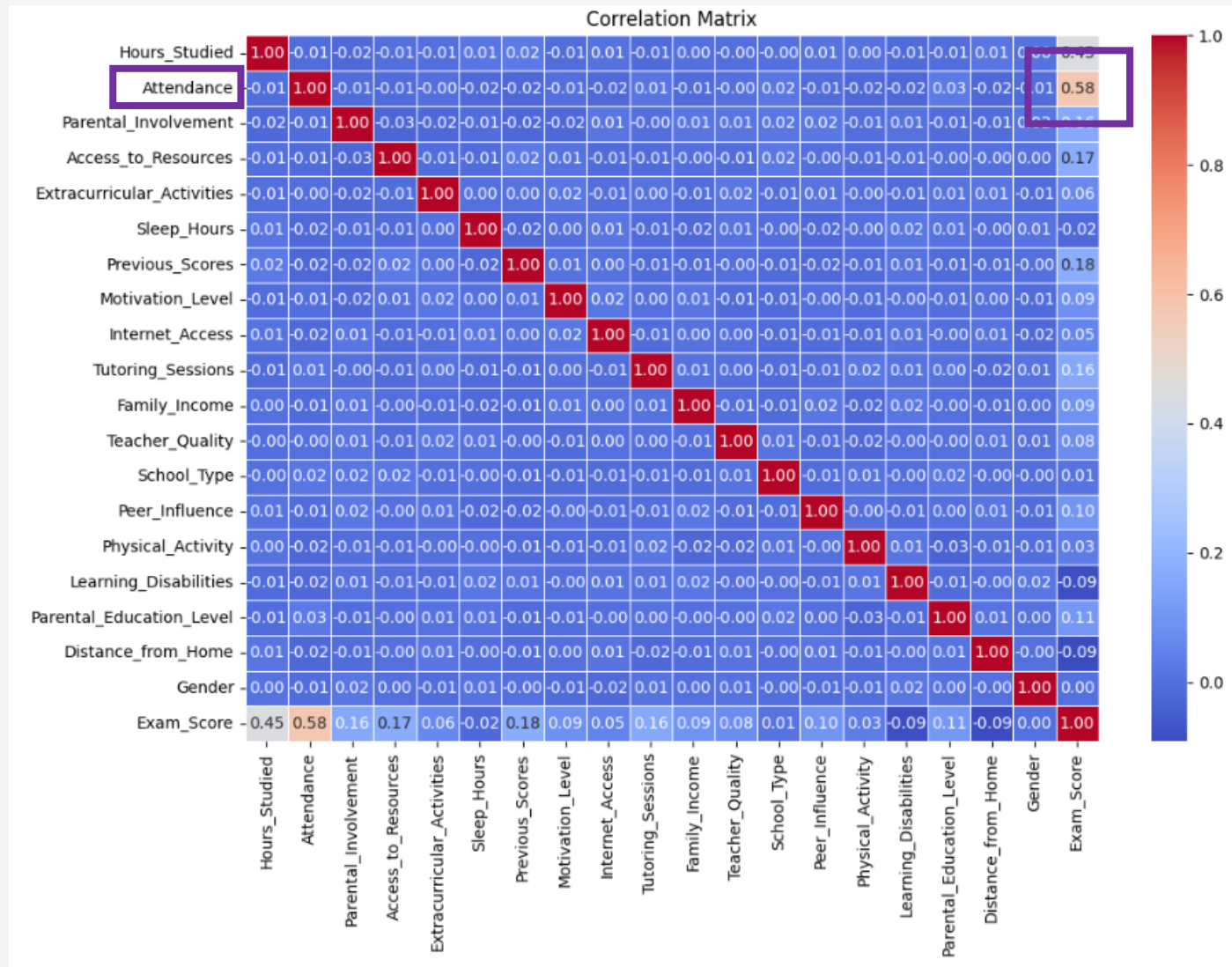
3 **Gender**

4 **Extracurricular Activities**

5 **Hours Studied**

6 **Previous Score**

Winner



A little Math (only to show the calculations)

STR(INT((COUNT(ID)+(COUNT(ID)*ABS(AVG([Gender Numeric]))))/2))+ ' of ' +STR(COUNT(ID))

When considering the men it is negative-> necessary the absolute value

$$x = N - x + Avg(Gender\ Numeric) * N$$

Total number of students in that grade group

$$x = \frac{N + Avg(Gender\ Numeric) * N}{2}$$

Example (Find Total Number Of Women in the grade group ≥ 98):

number of women with grade ≥ 98)

$$x = N - x + Avg(Gender\ Numeric) * N$$

number of men
in the group
grade ≥ 98

percentage of women more than men in that grade group

